

Athena AI

Enterprise-grade cybersecurity AI assistant — project introduction

Athena AI is a cybersecurity-focused LLM platform built around a fine-tuned base model, retrieval-augmented generation over a large security knowledge base, and a layer of supporting tools for research and reporting workflows. It targets security researchers, penetration testers, SOC analysts, and students. Repository: github.com/vrunalp199/Athena-AI (author: Vrunal Patil).

Core Model & Retrieval Stack

Component	Technology	Role
Base model	Qwen3.6-27B	Core language model
Fine-tuning	QLoRA (via Unsloth)	Cybersecurity-domain adaptation
Quantization	4-bit, BF16 precision	Fits training/inference on limited VRAM
Embeddings	BAAI/bge-large-en-v1.5	Text embedding for retrieval
Vector DB	Qdrant	Semantic / similarity search
Backend	Python + Flask	API and app serving
Storage	SQLite	Conversation / memory persistence

Tool Layer

Alongside the model and retrieval stack, Athena AI integrates a layer of external tools that extend it beyond pure Q&A into applied security-research workflows:

Tool / Module	Purpose
HexStrike	AI-driven orchestration layer for chaining offensive-security tools during authorized research and
report_generator.py	Turns findings and conversation context into structured security reports
log_analyzer.py	Parses and summarizes log data for anomaly / incident review
file_manager.py	Handles workspace file I/O for the assistant's tool calls
Qdrant client / retrieval.py	Executes embedding search against the vector knowledge base

Note: HexStrike itself is a separate open-source project — Athena AI's tool layer calls out to it rather than reimplementing it. It sits behind the assistant's reasoning loop the same way SEARCH_PDF and SEARCH_WEB actions do in the project's simplified variants — the model decides when a tool call is warranted rather than running it on every turn.

Knowledge Base & RAG Flow

The retrieval layer targets a 95+ GB corpus drawn from CVE databases, MITRE ATT&CK, security advisories, threat-intel reports, malware analysis writeups, and incident-response playbooks. A query is embedded, matched against Qdrant (cosine distance, top-K = 10, chunk size 1000 / overlap 150), and the retrieved context is assembled and passed to the Qwen3.6-27B model with the cybersecurity LoRA applied.

System Architecture

User → Athena AI Interface → Memory & Context Layer → [Qdrant Vector DB | Cyber LoRA | Tool Layer] → Qwen3.6-27B Base Model → Response

Training Infrastructure

Provider	RunPod
GPU	NVIDIA H100 SXM (80GB VRAM)
Framework	PyTorch + Unsloth, Flash Attention
OS / CUDA	Ubuntu 22.04, CUDA 12.x

Roadmap

- Phase 1 — Core AI assistant, memory system, prompt engineering
- Phase 2 — Cybersecurity LoRA, advanced RAG, vector search optimization
- Phase 3 — Multi-agent architecture, threat-intelligence agent, malware-analysis agent
- Phase 4 — SIEM integration, security automation, enterprise deployment

Intended Use

Per the project's own README: Athena AI is intended for educational purposes, cybersecurity research, defensive security operations, and authorized security testing only. Users are responsible for complying with all applicable laws and regulations. This applies equally to any integrated offensive-tooling layer such as HexStrike — it is scoped to authorized, consented testing engagements, not unauthorized access.